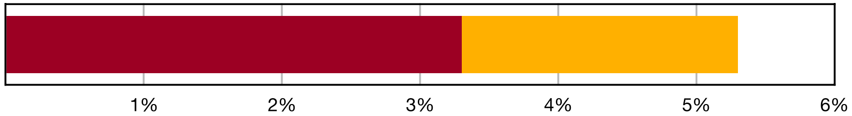
ADVANCED
Stage 6

Functions (Adv), F2 Graphing (Adv)
Non-Calculus Graphing (Y12)

Teacher: SHS Maths
Exam Equivalent Time: 78 minutes *(based on allocation of 1.5 minutes per mark)*

F2 Graphing



Transformations
Non-Calculus Graphing

*SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.

HISTORICAL CONTRIBUTION

- *F2 Graphing Techniques* has contributed an average of 5.3% per Advanced exam since the new syllabus was introduced in 2020.
- We have split the topic into 2 categories for analysis purposes: *1-Transformations (3.3%) and 2-Non-Calculus Graphing (2.0%).*
- This analysis looks at *Non-Calculus Graphing.*

HSC ANALYSIS - What to expect and common pitfalls

- *Non-Calculus Graphing* was last examined in 2023 in a 4-mark question that required students to graph two simple quadratics, find their points of intersection and solve an associated inequality equation.
- The most common question type is easily the graphing of quotient functions that require students to find intercepts and both vertical and horizontal asymptotes. This was last examined in the 2021 exam and relatively well answered.
- The old *Ext1* course provides a number of past HSC examples that have been recategorised into this *Advanced* topic. These will prove challenging and provide high quality revision. They are identified by the inclusion of ADV' in their title.
- The algebraic manipulation of some quotient functions before graphing can save significant time. We recommend reviewing *F2 EQ-Bank 11* to illustrate this.
- Students should be looking for the possibility of functions being odd or even. When identified, graphing solutions can be produced much more efficiently *(see 2ADV' F2 2012 HSC 13b).*

Questions

1. Functions, 2ADV F2 SM-Bank 9 MC

The graph of the function $f(x) = \dfrac{3x+2}{5-x}$, has asymptotes at

A. $x = -5,\ y = \dfrac{3}{2}$

B. $x = \dfrac{2}{3},\ y = -3$

C. $x = 5,\ y = 3$

D. $x = 5,\ y = -3$

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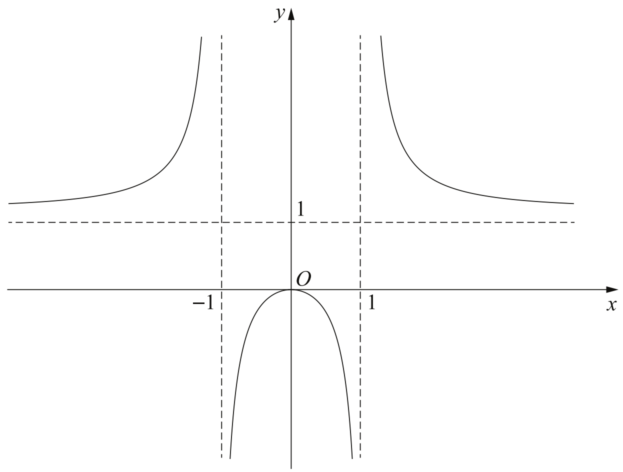
2. Functions, 2ADV' F2 2019 HSC 4 MC

The diagram shows the graph of $y = f(x)$.



Which equation best describes the graph?

A. $y = \dfrac{x}{x^2-1}$

B. $y = \dfrac{x^2}{x^2-1}$

C. $y = \dfrac{x}{1-x^2}$

D. $y = \dfrac{x^2}{1-x^2}$

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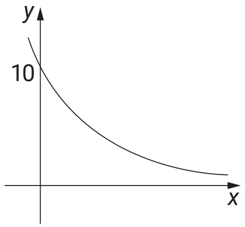
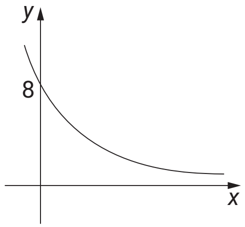
3. Functions, 2ADV F2 2021 HSC 5 MC

Which of the following best represents the graph of $y = 10(0.8)^x$?

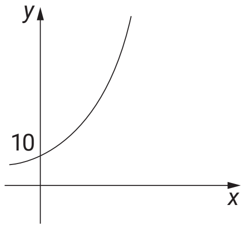
A.



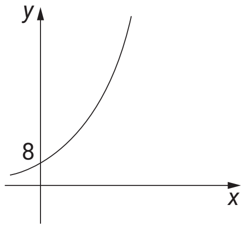
B.



C.



D.



4. Functions, 2ADV F2 SM-Bank 10 MC

The graph of the function $f(x) = \dfrac{x-3}{2-x}$ has asymptotes

A. $x = 3, \quad y = 2$

B. $x = -2, \quad y = 1$

C. $x = 2, \quad y = -1$

D. $x = 2, \quad y = 1$

5. Functions, 2ADV' F2 2015 HSC 5 MC

What are the asymptotes of $y = \dfrac{3x}{(x+1)(x+2)}$

A. $y = 0, \quad x = -1, \quad x = -2$

B. $y = 0, \quad x = 1, \quad x = 2$

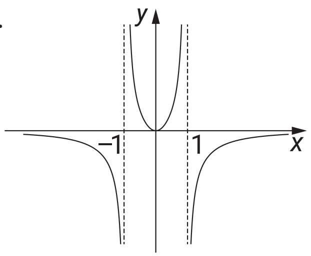
C. $y = 3, \quad x = -1, \quad x = -2$

D. $y = 3, \quad x = 1, \quad x = 2$

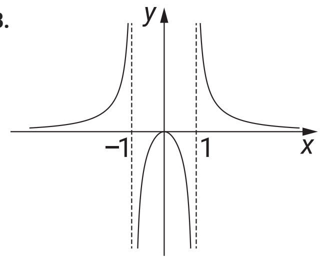
6. Functions, 2ADV' F2 2017 HSC 5 MC

Which graph best represents the function $y = \dfrac{2x^2}{1-x^2}$?

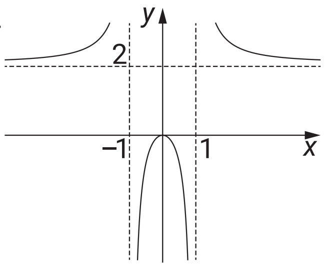
A.



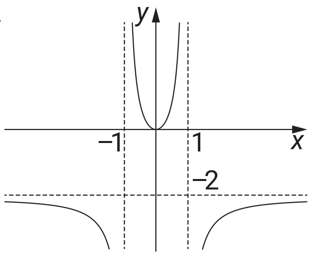
B.



C.



D.



7. Functions, 2ADV F2 2022 SPEC2 3 MC

The graph of $y = \dfrac{x^2 + 2x + c}{x^2 - 4}$, where $c \in R$, will always have

A. two vertical asymptotes and one horizontal asymptote.

B. a vertical asymptote with equation $x = -2$ and one horizontal asymptote with equation $y = 1$.

C. one horizontal asymptote with equation $y = 1$ and only one vertical asymptote with equation $x = 2$
.

D. a horizontal asymptote with equation $y = 1$ and at least one vertical asymptote.

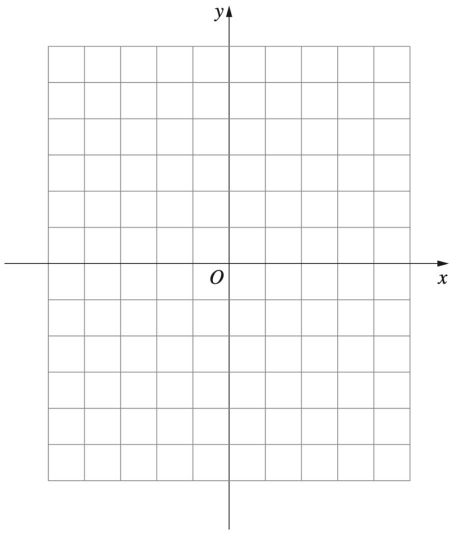
8. Functions, 2ADV F1 SM-Bank 35

i. Sketch the function $y = f(x)$ where $f(x) = (x-1)^3$ on a number plane, labelling all intercepts. ***(1 mark)***

ii. On the same graph, sketch $y = -f(-x)$. Label all intercepts. ***(2 marks)***

9. Functions, 2ADV F2 2023 HSC 19

a. Sketch the graphs of the functions $f(x) = x - 1$ and $g(x) = (1 - x)(3 + x)$ showing the x-intercepts. ***(2 marks)***



b. Hence, or otherwise, solve the inequality $x - 1 < (1 - x)(3 + x)$. ***(2 marks)***

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10. Functions, 2ADV F2 EQ-Bank 9

Consider the function $f(x) = \dfrac{1}{4x - 1}$.

i. Find the domain of $f(x)$. ***(1 mark)***

ii. Sketch $f(x)$, showing all asymptotes and intercepts? ***(2 marks)***

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11. Functions, 2ADV' F2 2012 HSC 13b

i. Find the horizontal asymptote of the graph $y = \dfrac{2x^2}{x^2 + 9}$. ***(1 mark)***

ii. Without the use of calculus, sketch the graph $y = \dfrac{2x^2}{x^2 + 9}$, showing the asymptote found in part (i). ***(2 marks)***

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12. Functions, 2ADV F2 2021 HSC 19

Without using calculus, sketch the graph of $y = 2 + \dfrac{1}{x + 4}$, showing the asymptotes and the x and y intercepts. ***(3 marks)***

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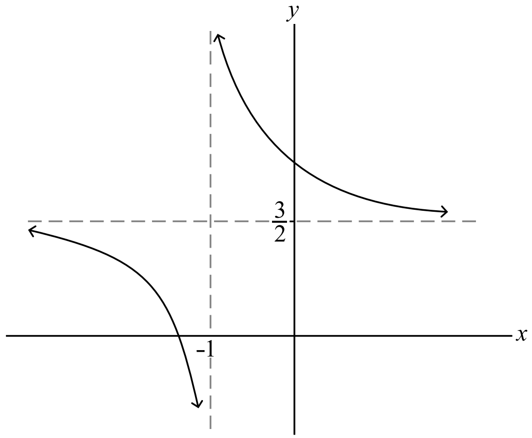
13. Functions, 2ADV F2 EQ-Bank 2

A graph of the hyperbola $y = \dfrac{1}{x + p} + q$ is shown, where p and q are constants.



Find the values of p and q and hence the graphs x-axis intercept. ***(2 marks)***

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14. Functions, 2ADV F2 SM-Bank 13

i. Show that the function $y = \dfrac{1 - e^x}{1 + e^x}$ is an odd function? ***(1 mark)***

ii. Sketch $y = \dfrac{1 - e^x}{1 + e^x}$, labelling all intercepts and asymptotes. ***(2 marks)***

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15. Functions, 2ADV F2 SM-Bank 10

Consider the function $f(x) = \dfrac{x}{4 - x^2}$.

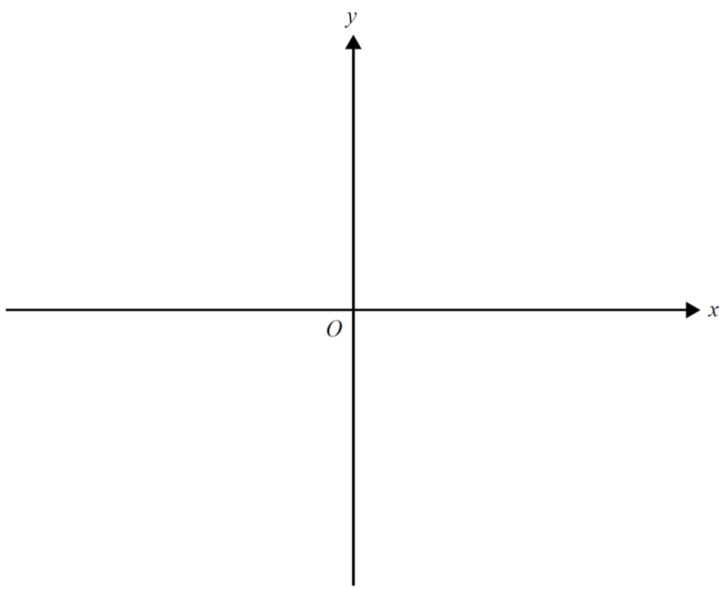
i. Identify the domain of $f(x)$. ***(1 mark)***

ii. Sketch the graph $y = f(x)$, showing all intercepts and asymptotes. ***(3 marks)***

16. Functions, 2ADV F2 EQ-Bank 11

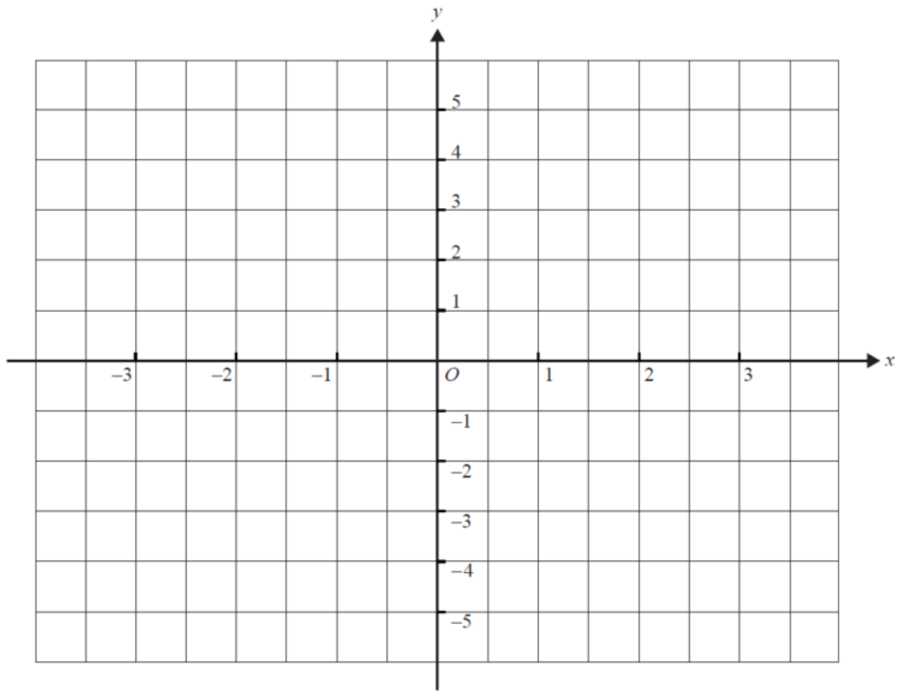
On the axes below, sketch the graph of $f(x)=\dfrac{2x-2}{x+1}$.

Label all axis intercepts. Label each asymptote with its equation. ***(4 marks)***



17. Functions, 2ADV F2 SM-Bank 12

Sketch the graph of $f(x)=\dfrac{2x+1}{x-1}$. Label the axis intercepts with their coordinates and label any asymptotes with the appropriate equation. ***(4 marks)***



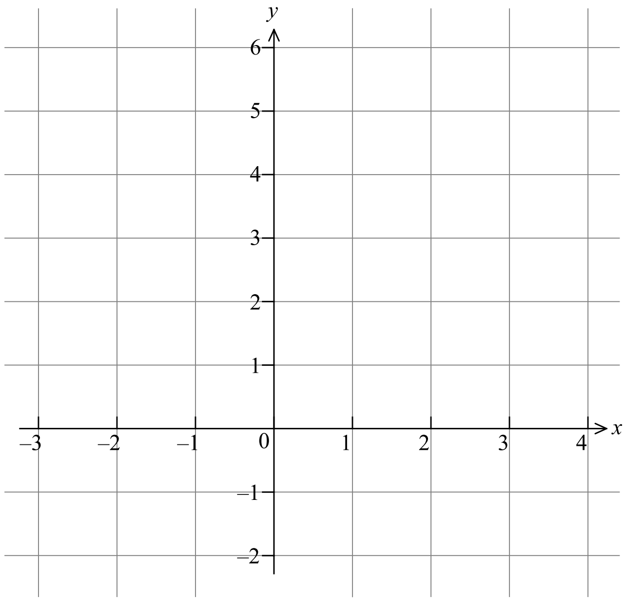
18. Functions, 2ADV' F2 2007 HSC 3b

i. Find the vertical and horizontal asymptotes of the hyperbola $y=\dfrac{x-2}{x-4}$ and hence sketch the graph of $y=\dfrac{x-2}{x-4}$. ***(3 marks)***

ii. Hence, or otherwise, find the values of x for which $\dfrac{x-2}{x-4}\leq 3$. ***(2 marks)***

19. Functions, 2ADV F2 2023 MET1 3

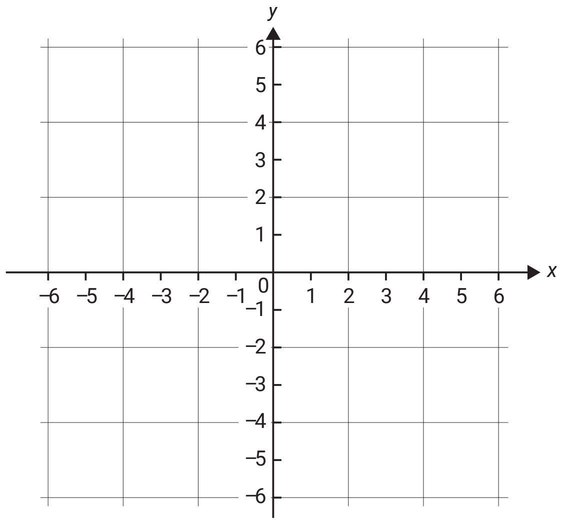
a. Sketch the graph of $f(x) = 2 - \dfrac{3}{x-1}$ on the axes below, labelling all asymptotes with their equation and axial intercepts with their coordinates. ***(3 marks)***



b. Find the values of x for which $f(x) \leq 1$. ***(1 mark)***

20. Functions, 2ADV F2 SM-Bank 20

a. Sketch the graph of $y = 1 - \dfrac{2}{x-2}$ on the axes below. Label asymptotes with their equations and axis intercepts with their coordinates. ***(3 marks)***



b. Find the values of x for which $1 - \dfrac{2}{x-2} \geq 3$. ***(1 mark)***

Worked Solutions

1. Functions, 2ADV F2 SM-Bank 9 MC

$$\begin{aligned} f(x) &= \frac{3x+2}{5-x} \\ &= \frac{-(15-3x)+17}{5-x} \\ &= -3+\frac{17}{5-x} \end{aligned}$$

Vertical asymptote: $x=5$
Horizontal asymptote: $y=-3$
$\Rightarrow D$

2. Functions, 2ADV' F2 2019 HSC 4 MC

By elimination:
Graph is an even function
$\Rightarrow f(x)=f(-x)$
$\therefore$ Eliminate A and C

When $-1<x<1,\ \ y\leq 0$
$\therefore$ Eliminate D
$\Rightarrow B$

3. Functions, 2ADV F2 2021 HSC 5 MC

By elimination:
When $x=0\ ,\ y=10(0.8)^0=10$
$\rightarrow$ Eliminate B and D
As $x\rightarrow\infty,\ y\rightarrow 0$
$\rightarrow$ Eliminate C
$\Rightarrow A$

4. Functions, 2ADV F2 SM-Bank 10 MC

$$\begin{aligned} y &= -\left(\frac{3-x}{2-x}\right) \\ &= -\left(\frac{2-x}{2-x}+\frac{1}{2-x}\right) \\ &= -1-\frac{1}{2-x} \end{aligned}$$

$\therefore$ Asymptotes: $x=2,\ y=-1$
$\Rightarrow\ C$

Worked Solutions

5. Functions, 2ADV' F2 2015 HSC 5 MC

$$y=\frac{3x}{(x+1)(x+2)}$$

Asymptotes at $x=-1$ and $x=-2$
As $x\rightarrow\infty, y\rightarrow 0^+$
As $x\rightarrow -\infty, y\rightarrow 0^-$
$\therefore$ Horizontal asymptote at $y=0$
$\Rightarrow A$

6. Functions, 2ADV' F2 2017 HSC 5 MC

$$\begin{aligned} y &= \frac{2x^2}{(1-x^2)} \\ &= -\frac{(2-2x^2-2)}{(1-x^2)} \\ &= -\frac{2(1-x^2)}{(1-x^2)}-\frac{2}{(1-x^2)} \\ &= -2-\frac{2}{(1-x^2)} \end{aligned}$$

As $x\rightarrow\infty,\ \ y\rightarrow -2$
$\therefore$ Horizontal asymptote at $y=-2$
$\Rightarrow D$

7. Functions, 2ADV F2 2022 SPEC2 3 MC

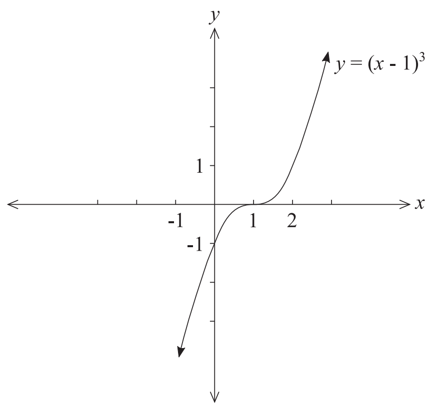
$$y=\frac{x^2+2x+c}{x^2-4}\ \ \Rightarrow\ \ y=\frac{x^2+2x+c}{(x-2)(x+2)}$$

Vertical asymptotes: $x=2,\ x=-2$
$\rightarrow$ However, if $c=0,$ only 1 vertical asymptote at $x=2.$
Horizontal asymptote: $y=1$
$\Rightarrow D$

♦♦ Mean mark 38%.
41% of students chose A and did not consider when $c=0$

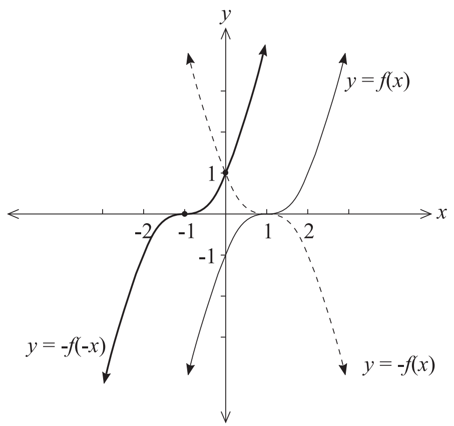
8. Functions, 2ADV F1 SM-Bank 35

i. $y=(x-1)^3 \Rightarrow y=x^3$ shifted 1 unit to the right.



ii. $y=-f(x) \;\Rightarrow$ reflect $y=(x-1)^3$ in x-axis.

$y=-f(-x) \;\Rightarrow$ reflect $y=-f(x)$ in y-axis.



9. Functions, 2ADV F2 2023 HSC 19

a. $g(x)$ cuts x-axis at 1 and -3.

$g(x)_{\text{max}}=g(-1)=4$

Find intersection of graphs:

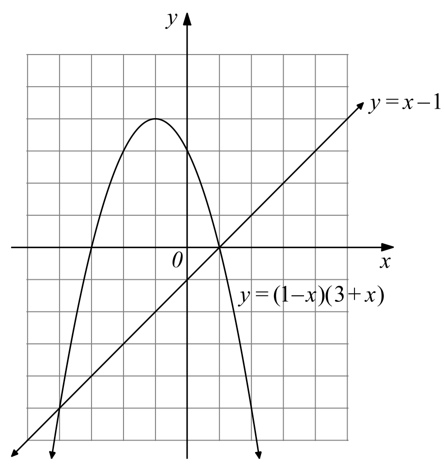
$$(1-x)(3+x)=x-1$$

$$3+x-3x-x^2x-1$$

$$x^2+3x-4=0$$

$$(x+4)(x-1)=0$$

Intersections at: $(1,0),(-4,-5)$



b. From the graph:

$x-1<(1-x)(3+x)$ when $-4<x<1$

Test $x=0$:

$0-1<(1-0)(3+0) \;\Rightarrow\; -1<3$ (correct)

10. Functions, 2ADV F2 EQ-Bank 9

i. $4x - 1 \neq 0$

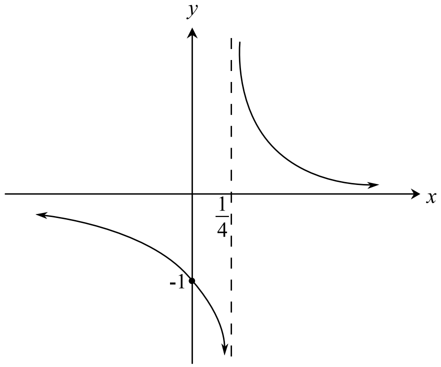
$$x \neq \frac{1}{4}$$

$$\therefore \text{ Domain: } \left\{\text{all real } x,\ x \neq \frac{1}{4}\right\}$$

ii. When $x = 0,\ y = -1$

As $x \to \infty,\ y \to 0^+$

As $x \to -\infty,\ y \to 0^-$



11. Functions, 2ADV' F2 2012 HSC 13b

i.
$$y = \frac{2x^2}{x^2+9} = \frac{2}{1 + \frac{9}{x^2}}$$

As $x \to \infty,\ y \to 2$

As $x \to -\infty,\ y \to 2$

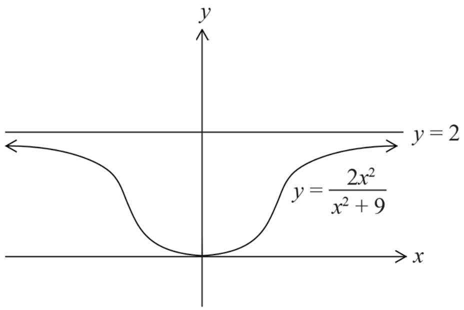
$\therefore$ Horizontal asymptote at $y = 2$

ii. At $x = 0,\ y = 0$

$$f(x) = \frac{2x^2}{x^2+9} \geq 0 \text{ for all } x$$

$$f(-x) = \frac{2(-x)^2}{(-x)^2+9} = \frac{2x^2}{x^2+9} = f(x)$$

Since $f(x) = f(-x) \Rightarrow$ EVEN function



12. Functions, 2ADV F2 2021 HSC 19

Asymptotes: $x = -4$

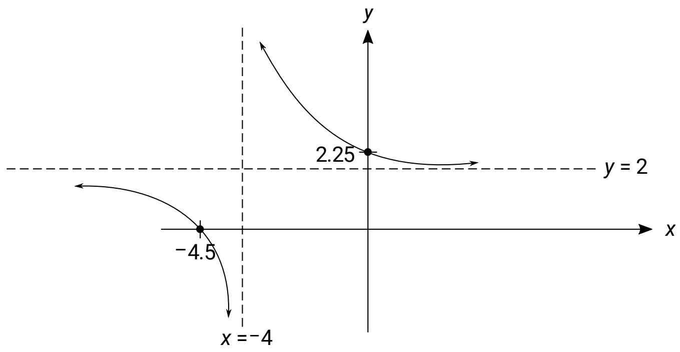
As $x \to \infty, y \to 2$

y-intercept occurs when $x = 0$:

$y = 2.25$

x-intercept occurs when $y = 0$:

$$2 + \frac{1}{x+4} = 0 \ \Rightarrow \ x = -4.5$$



13. Functions, 2ADV F2 EQ-Bank 2

Vertical asymptote at $x = -1 \ \Rightarrow p = 1$

Horizontal asymptote at $y = \frac{3}{2} \ \Rightarrow q = \frac{3}{2}$

$$y = \frac{1}{x+1} + \frac{3}{2}$$

x-intercept when $y = 0$:

$$\begin{aligned} \frac{1}{x+1} + \frac{3}{2} &= 0 \\ \frac{1}{x+1} &= -\frac{3}{2} \\ 3x + 3 &= -2 \\ 3x &= -5 \\ x &= -\frac{5}{3} \end{aligned}$$

14. Functions, 2ADV F2 SM-Bank 13

i. $f(x) = \dfrac{1-e^x}{1+e^x}$

$$\begin{aligned} f(-x) &= \frac{1-e^{-x}}{1+e^{-x}} \times \frac{e^x}{e^x} \\ &= \frac{e^x-1}{e^x+1} \\ &= -\frac{1-e^x}{1+e^x} \\ &= -f(x) \end{aligned}$$

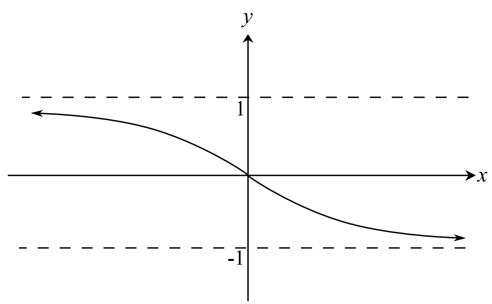
$\therefore f(x)$ is ODD.

ii. $y = \dfrac{1-e^x}{1+e^x} \times \dfrac{e^{-x}}{e^{-x}} = \dfrac{e^{-x}-1}{e^{-x}+1} = 1 - \dfrac{2}{e^{-x}+1}$

As $x \to \infty$, $\dfrac{2}{e^{-x}+1} \to 2$, $y \to -1$

As $x \to -\infty$, $\dfrac{2}{e^{-x}+1} \to 0$, $y \to 1$

When $x = 0$, $y = 0$



15. Functions, 2ADV F2 SM-Bank 10

i. $4-x^2 \neq 0$

$\Rightarrow x \neq 2 \text{ or } -2$

$\therefore \text{ Domain: all } x,\ x \neq \pm 2$

ii. Asymptotes at $x = \pm 2$

Passes through $(0,0)$

As $x \to 2^-,\ y \to \infty$

$x \to 2^+,\ y \to -\infty$

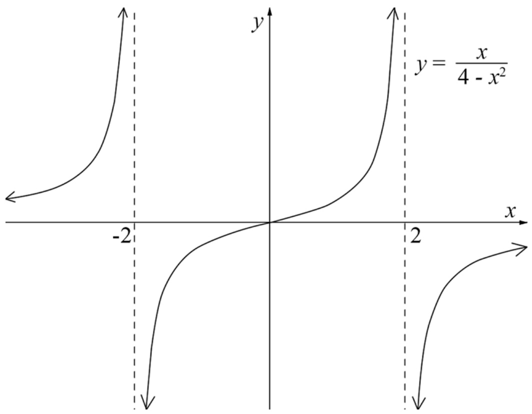
As $x \to -2^-,\ y \to \infty$

$x \to -2^+,\ y \to -\infty$

As $x \to \infty,\ y \to 0$

$x \to -\infty,\ y \to 0$

COMMENT: Note that $x \to 2^-$ is a short way of writing as x approaches 2 from the negative (or left-hand) side. This notation can save time when required.



16. Functions, 2ADV F2 EQ-Bank 11

$$\frac{2x-2}{x+1} = \frac{2(x+1)-4}{x+1}$$

$$= 2 - \frac{4}{(x+1)}$$

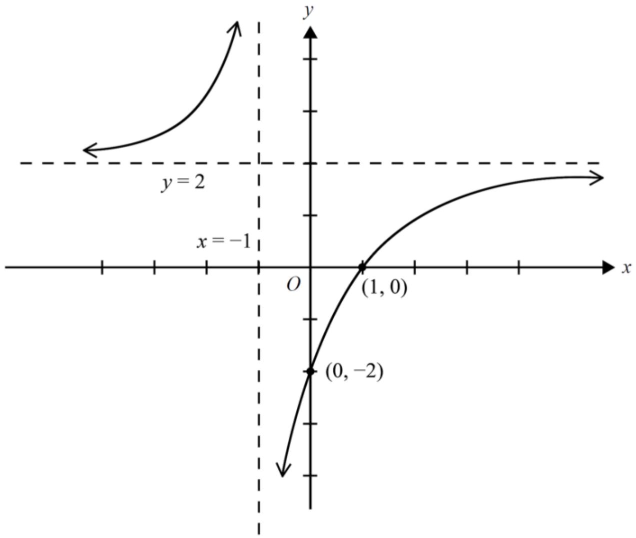
Asymptotes: $x = -1,\ y = 2$

As $x \to \infty,\ y \to 2(-)$

As $x \to -\infty,\ y \to 2(+)$

As $x \to -1(-),\ y \to \infty$

As $x \to -1(+),\ y \to -\infty$



17. Functions, 2ADV F2 SM-Bank 12

COMMENT: Manipulation of the equation makes graphing much easier.

$$\begin{aligned}\frac{2x+1}{x-1} &= \frac{2x-2+3}{x-1}\\ &= \frac{2(x-1)+3}{x-1}\\ &= 2+\frac{3}{x-1}\end{aligned}$$

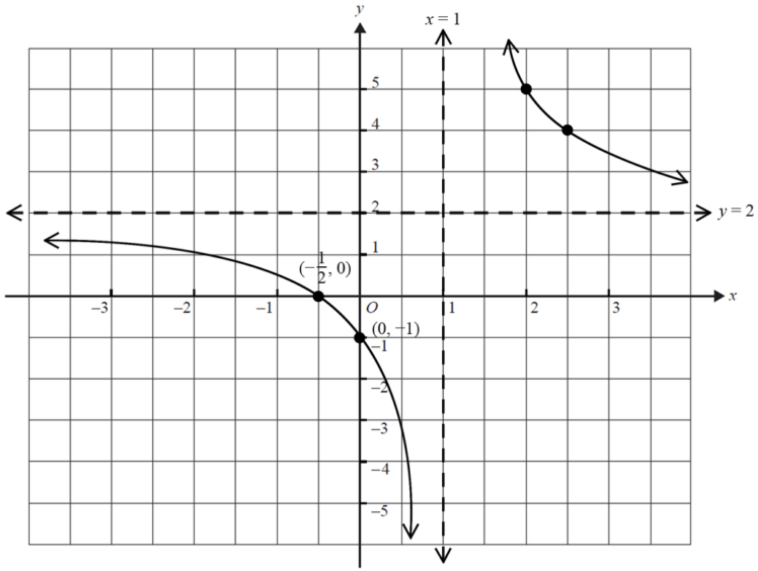
Asymptotes: $x=1,\ \ y=2$

As $x\to\infty,\ \ y\to 2(+)$

As $x\to-\infty,\ \ y\to 2(-)$

As $x\to-1(-),\ \ y\to-\infty$

As $x\to-1(+),\ \ y\to\infty$



18. Functions, 2ADV' F2 2007 HSC 3b

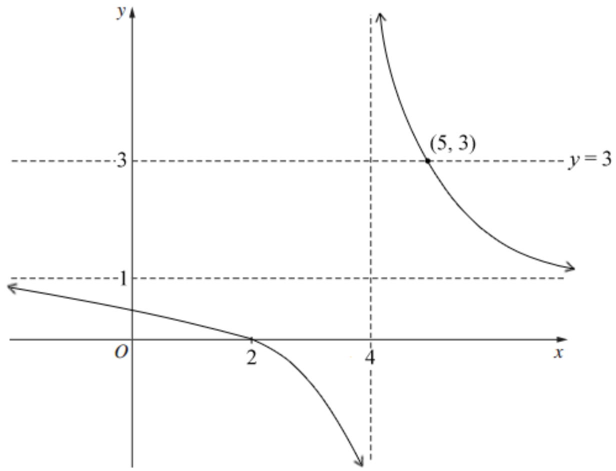
i. $y=\dfrac{x-2}{x-4}$

Vertical asymptote at $x=4$

$$\lim_{x\to\infty}\frac{x-2}{x-4}=\lim_{x\to\infty}\frac{1-\frac{2}{x}}{1-\frac{4}{x}}=1$$

y–intercept $=\dfrac{1}{2}$

x–intercept $=2$



ii. Find x so that $\dfrac{x-2}{x-4}\le 3$

$$\begin{aligned}\frac{x-2}{x-4}&=3\\ x-2&=3x-12\\ 2x&=10\\ x&=5\end{aligned}$$

$\Rightarrow (5,3)$ is the intersection of $y=3$ and $y=\dfrac{x-2}{x-4}$

$\therefore\ \dfrac{x-2}{x-4}\le 3$ when $x<4$ and $x\ge 5$.

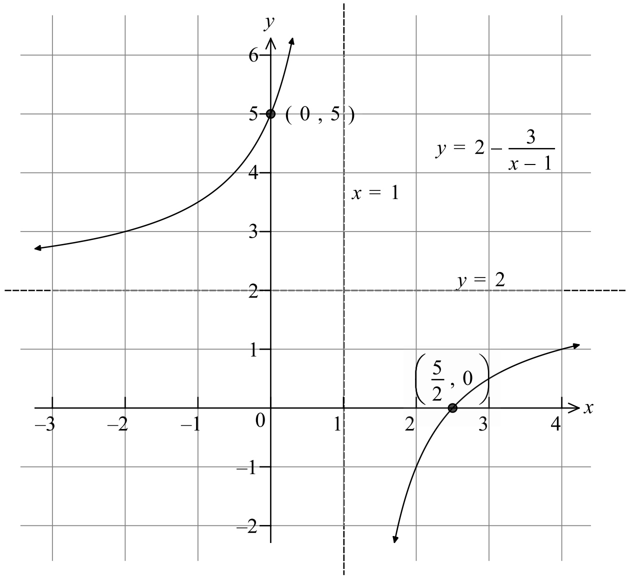
19. Functions, 2ADV F2 2023 MET1 3

a. Vertical asymptote when $x = 1$

y-int: $y = 2 - (-3) \;\Rightarrow\; y = 5$

x-int: $2 - \dfrac{3}{x-1} = 0 \;\Rightarrow\; x = \dfrac{5}{2}$

As $x \to \infty,\; y \to 2^-;\; x \to -\infty,\; y \to 2^+$



b. From the graph:

$f(x) = 1$ when $x = 4$

$x > 1$ to the right of the vertical asymptote

$\therefore\; f(x) \leq 1$ when $1 < x \leq 4$ or $(1, 4]$

♦♦ Mean mark (b) 38%.
MARKER'S COMMENT:
Many students did not use their graph from part (a).

20. Functions, 2ADV F2 SM-Bank 20

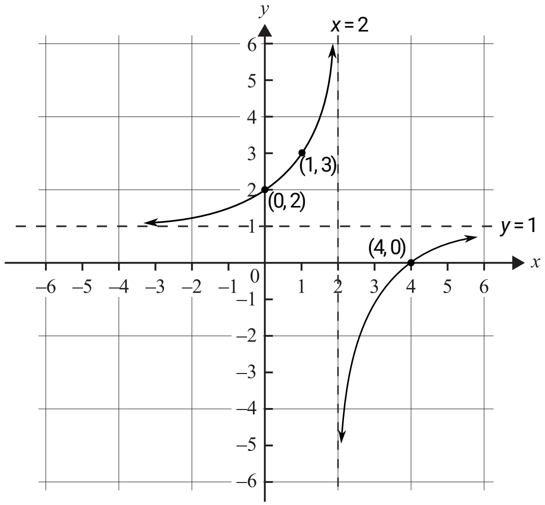
a. Asymptotes:

$x = 2$

As $x \to \pm\infty,\; y \to 1 \;\Rightarrow$ Asymptote at $y = 1$

y-intercept at $(0, 2)$

x-intercept at $(4, 0)$



b. By inspection of the graph:

$1 - \dfrac{2}{x-2} \geq 3$ for $x \in [1, 2)$